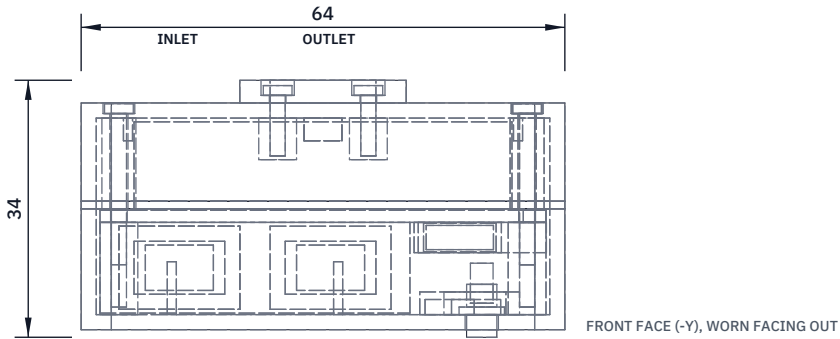


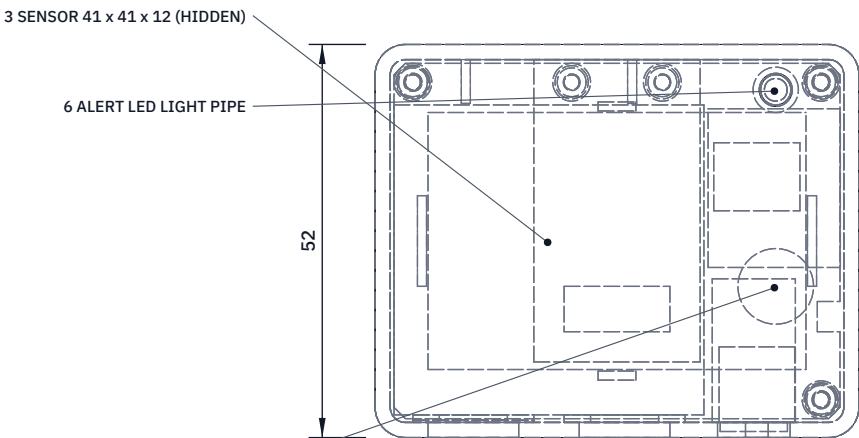
CONCEPT, NOT FOR FABRICATION

DustBadge · github.com/BoujeeEnjinia1701/dustbadge



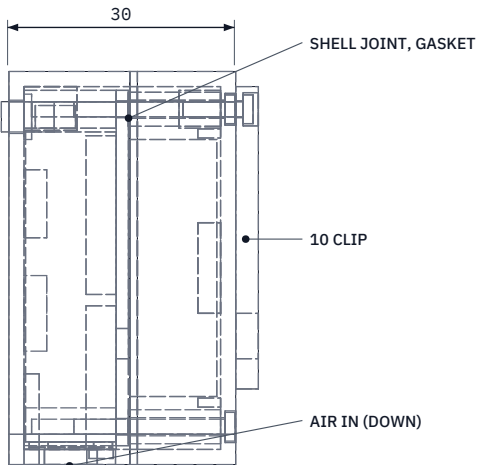
TOP VIEW

Scale 1:1; looking down (along -Z)



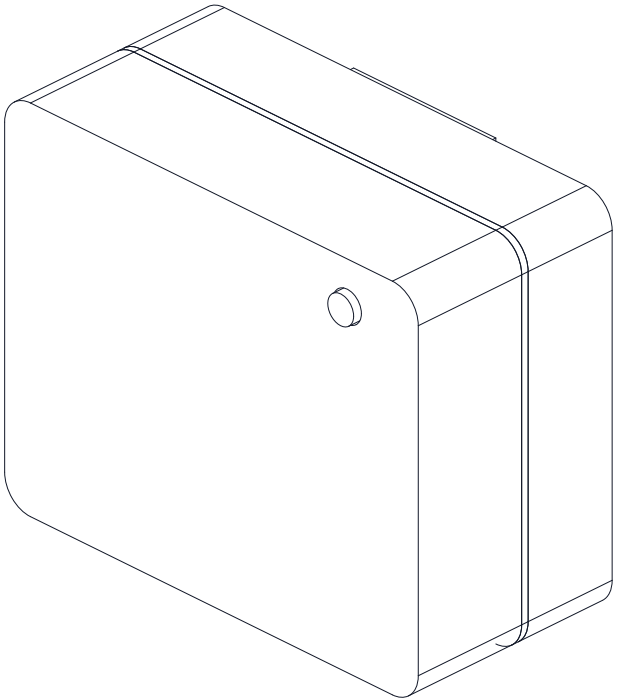
FRONT VIEW

Scale 1:1; looking at the front (along +Y)



RIGHT VIEW

Scale 1:1; looking at the right side (along -X)



ISOMETRIC VIEW

Not to scale; seen from the front right and above, about 30 deg elevation

MAIN DIMENSIONS AND INTERFACES (MM)

Shells 64 x 52 x 30; 34 deep with clip; wall 2; gasket 1 at 14 from rear
Sensor SPS30 class 41 x 41 x 12, ports down onto two slots
Slots 12 x 7: inlet (screened) at X -19, outlet at X 1
USB-C opening at X 18; humidity vent at X 18; all in the bottom face
Cell 50 x 34 x 11.5, 2,000 mAh, protected, with thermistor
Board 59 x 47 clamped by three M2 x 20 screws; TPU gasket and port seals
Mass 119.9 g (DBG-CAL-001); inlet 242 from nose and mouth when worn
Not intrinsically safe; not for gassy mines or explosive atmospheres
Wear shaded when ambient is above 40 °C in full sun
Third-angle; front view from -Y; origin at the badge center, rear face Y = 0

DUSTBADGE

github.com/BoujeeEnjinia1701/dustbadge

General arrangement

DRAWING NO. DBG-DWG-001			REV P4	SHEET 1 of 1	SIZE ANSI B
SCALE 1:1	UNITS mm	PROJECTION 	DATE 2026-09-30	DRAWN Amish Chadha	

MATERIAL / NOTES
PETG shells; bought-in parts per bom/bom.csv. PRELIMINARY, NOT FOR FABRICATION